

# Social and Economic Activity Predicts Daily Hospital Emergency Visits <sup>††</sup>

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## Abstract

We show a strong effect of social and economic activity on emergency room utilization using daily data from the universe of healthcare facilities in Chile, and novel measures of activity based on mobility. We examine the onset of the COVID-19 pandemic as a case-study to show that the pre-pandemic estimated relationship between mobility and emergencies is able to explain a large portion of the 50 percent drop emergency visits in the country. Our results highlight that novel mobility data can be used to improve high-frequency predictions of emergency room demand.

**Keywords:** Health Care, Emergencies, Mobility.

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# 1 Introduction

Emergency department (ED) visits are the portal of entry for roughly half of all hospital admissions in the U.S. (Marcozzi et al., 2018; Pitts et al., 2010). As the ED is a crucial component of health care delivery, it poses significant challenges for efficiency and quality (Bernstein et al., 2009). Understanding the factors that drive ED utilization is essential for predicting the demand for ED and for designing effective policies and interventions to improve health care performance.

In this paper, we show that daily measures of social and economic activity are highly predictive of daily demand for ED services. Previous theoretical and empirical studies have discussed many potential links between economic factors and health. Our key contribution is to leverage novel daily proxies of social and economic activity from mobility data to provide empirical evidence that relates activity to daily demand for ED. To that aim, we use detailed administrative data for the universe of providers with an ED in Chile combined with novel data on mobility recently released by Google.

We demonstrate the potential use of the model using the COVID-19 pandemic as a case study. The pandemic caused a sudden and large drop in economic and social activity due to strict lock-down policies and changes in individual behavior. Concurrently, emergency rooms in Chile and worldwide saw an unprecedented plunge in demand (Garcia et al., 2020; Rodríguez-Leora et al., 2020).<sup>1</sup> Our model, estimated only with data from *before* the pandemic as the training dataset, accurately predicts the drop in ED demand as a consequence of the drop in mobility. This is a particularly challenging out-of-sample prediction as the mobility patterns post-pandemic are outside the support of the pre-pandemic patterns used to estimate the model. Additionally, the health crisis may have affected the underlying relationship between mobility and ED visits in many ways, for reasons including an increased reluctance to visit hospitals out of fear of contagion. Still, our model can predict almost all of the decrease in ED demand observed in the first months of the pandemic.

Overall, our results show a strong link between activity and ED demand and the potential of using newly-available high-frequency activity data to explain and forecast ED demand. We provide two additional analyses that provide empirical support for this link. First, we leverage our daily data to show that visits related to trauma and poisoning are higher during working days than on the weekends and that this relationship disappears during the summer months. Consistent with our earlier findings, we also show that this intra-week

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<sup>1</sup>For instance, according to NHS England figures, in March 2020 the number of people attending Accident and Emergency departments in English hospitals was down 29 percent from the same month the previous year. Emergency admissions were also down, falling by 23 percent on last March, to nearly 428,000 (“Coronavirus: A&E visits drop sharply as calls to 111 ‘double’,” Philippa Roxby, BBC News, April 9, 2020).

pattern disappeared during the pandemic. Additionally, we found no discernible increases in mortality at the onset of the pandemic, in spite of the almost 50% drop in ED demand. This is also consistent with a lower incidence of emergencies attributed to lower activity.

Our paper contributes to studies that analyze and forecast the demand for ED (Marcilio et al. 2013; Calegari et al. 2016; Whitt and Zhang 2019; Hu et al. 2021; Bertsimas et al. 2022). We complement these studies by providing evidence that shows a strong link between demand and social and economic activity. Data on high-frequency mobility was previously hard to obtain, but many companies have released proprietary mobility data to the public.<sup>2</sup> Our results show that mobility data can be used as a key input for predictive models of daily demand for ED services.

We also contribute to the discussion analyzing the extent to which visits to the ED are actually urgent (Durand et al., 2011; Uscher-Pines et al., 2013). In fact, several studies have argued that many ED visits are actually non-urgent, and therefore could be handled with regular physician visits without any deterioration in the patient’s condition (see, e.g., Courtemanche et al., 2019).<sup>3</sup> The large drop in ED visits at the onset of the pandemic might suggest that a large fraction of pre-COVID-19 emergency-room visits were only marginally beneficial to patients. However, our findings suggest that the lower incidence of conditions due to lower exposure to risk played an important part in this decrease, bounding the role played by behavioral responses related to the decision to go to the ED.

Our paper relates to studies linking changes in economic conditions to health outcomes, starting with Ruhm (2000). Although most of the early work confirmed the “pro-cyclical” behavior of mortality uncovered by Ruhm for the U.S. between 1972 and 1991, several other studies reach a different conclusion when using other outcomes, other countries, or more recent time periods (see, e.g., Cutler et al., 2002; Ruhm, 2015, and Stevens et al., 2015). The variety of empirical results likely reflect the myriad of mechanisms linking economic outcomes to health care outcomes (Catalano et al., 2011).<sup>4</sup> Our paper provides novel high-frequency

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<sup>2</sup>Coster et al. (2017) provides a recent review of the literature analyzing the factors that make people with a potentially urgent condition to choose ED over non-urgent care.

<sup>3</sup>The OECD reports that between 12 and 56 percent of ED visits in OECD countries are non-urgent (James et al., 2017) and these represented \$38 billion per year in the U.S. (Delaune and Everett, 2008). One potential explanation is that patient preferences for seeking emergency care are high because of the wide array of medical services that are always accessible (Durand et al., 2012). An important caveat is that there is not a clear consensus of what an urgent visit is. See Durand et al. (2011) for a review of the classification methods used in the health literature and their implications. Also, UnitedHealthcare Group estimates that 18 out of 27 million ED visits in the US are avoidable, representing US 32 billion of potential savings (see <https://www.unitedhealthgroup.com/content/dam/UHG/PDF/2019/UHG-Avoidable-ED-Visits.pdf>), and Agency for Healthcare Research and Quality (2018) reports that between 13 to 27% of ED visits are avoidable which would represent savings of USD 4.4 billion annually in the US.

<sup>4</sup>In fact, the economic theory (e.g., Grossman, 1972) as well as the medical literature provide hypotheses for either positive or negative effects of unfavorable economic conditions on health. For example, an economic

evidence that ED usage increases with social and economic activity when macroeconomic conditions are stable (as in the period before the pandemic) and therefore most of the confounding mechanisms at play in longer periods of time are absent. Closer to this paper in that regard is Evans and Moore (2012) who show that deaths decline before the first day of the month and spike afterwards, and provide evidence from auxiliary data to argue that this pattern follows a range of economic activities. An important contribution of our paper is to use comprehensive data on both socio-economic activity and health-related outcomes.

Finally, our paper is related to a growing literature using mobility data to address questions related to the management of public health. Ilin et al. (2021) show that publicly available data on human mobility—collected by Google, Facebook, and other providers—can be used to evaluate the effectiveness of non-pharmaceutical interventions in public health and forecast the spread of COVID-19. Along the same lines, Pepe et al. (2020); Farboodi et al. (2021); Valentí and Martínez-de-Albéniz (2021); Carranza et al. (2022), among others, use cellphone tracking data to depict the evolution of social activity during the pandemic. We contribute to this literature by showing the use of mobility data in predicting emergency department visits in the pre-pandemic period, which also highlights its use beyond those directly associated with tracking mobility and beyond the pandemic context.

## 2 Context and Data

Our main empirical analysis combines novel daily data from the Google Community Mobility Reports with daily administrative data on ED admissions to study the relation between the two. We use data for Chile starting on February 15, 2020, when the mobility data begins to be widely available. Within our data period we distinguish between two subperiods: (i) from February 15 to March 13, our “*training*” data set before the COVID crisis hit the country (the “pre-COVID-19” period); and (ii) from March 13 to April 28, the early period of the COVID-19 crisis in Chile, which constitutes our “*out-of-sample*” dataset that we use to examine the ability of our model to predict ED visits.<sup>5</sup>

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crisis can affect family income and alter nutrition by reducing food consumption, but also the individual’s decision to work and the time devoted for caregiving, fertility decisions, access to healthcare services, and public health spending.

<sup>5</sup>Even if the timing of some of the specific lockdown measures differed across regions, the mobility data shows a systematic break in mobility patterns across all regions after March 13, which justifies the use of a single date to define the pre-COVID-19 period across all regions. This evidence is consistent with Farboodi et al. (2021), who show that individuals in the U.S. reduced their activities during the COVID-19 pandemic before any legal restrictions on movement were implemented. In this paper, we do not take a stand regarding the causes behind the drop in mobility, which are likely a combination of the policies implemented to increase social distancing and individuals’ personal decision to stay at home. Finally, the main results of the paper are robust to the definition of the COVID-19 starting date. See Appendix A5.4

The Chilean context is well-suited for this analysis for three important reasons: First, detailed, high-quality ED data is available daily for the universe of ED visits in the country, which allows us to construct a high-frequency (daily) panel at the regional and type-of-emergency levels that we combine with activity data. Second, during the period of our analysis healthcare facilities operated below their capacity, even after the beginning of the pandemic. Excess capacity of hospitals and ED rooms helps us rule out unmet demand due to supply-side factors as an explanation for the decrease in ED visits.<sup>6</sup> Third, the drop in ED visits in Chile after COVID-19 was comparable in magnitude to the decrease reported elsewhere.

## 2.1 Mobility Data: The Google Community Mobility Reports

Our mobility data comes from the Google Community Mobility Reports. The data was released by Google in an effort to inform the debate around COVID-19. These mobility reports use mobile geo-locations to compute an index of the time spent by cellphone users in six different categories of places: retail and recreation, grocery and pharmacy, parks, transit stations, workplaces and residences. For a particular day and location, the reports show how visits and length of stay at different places differ with respect to a baseline. The baseline is the median value, for the corresponding day of the week, during the 5-week period between January 3 to February 6, 2020. The data is available at the daily level for all 16 regions of Chile.<sup>7</sup> The mobility data was captured among users of Android cellphones who had opted in to turn on their location services.<sup>8</sup> Although the Google Mobility data we use in this paper was released in the context of the COVID-19 pandemic, previous studies have shown that Google Location data has an accuracy level similar to a GPS (see an example of Ruktanonchai et al., 2018 for the UK). The mobility data has been used recently by Wellenius et al. (2021) and Dasgupta et al. (2020) to estimate the impact of state policies to foster social distancing on mobility during the pandemic, and by *The Economist* to construct an “economic activity” index for U.S. States.<sup>9</sup>

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<sup>6</sup>Hospital congestion due to a surge on respiratory ED visits associated with COVID-19 cases is not a concern during our period of analysis. In Appendix A1, we provide details on the evolution of COVID-19 cases and deaths between March 1 and April 28, 2020, and plots of the use of ICU beds and mechanical ventilators to show there health care system operated below its capacity during our study period.

<sup>7</sup>Regions are the main sub-national administrative units. We present in Appendix A2 plots of the six mobility indices, both in Chile and in the U.S. Comparing how the mobility indices changed with respect to those in the U.S. the mobility in Chile decreased in most categories much more than in the U.S., especially in retail and recreation, and in grocery and pharmacy.

<sup>8</sup>According to the 2017 Socio-Economic Survey (CASEN), 92 percent of Chileans over 18 have a working cellphone. As of January 2020, Android had a market share of 85 percent (see <https://gs.statcounter.com/os-market-share/mobile/chile>).

<sup>9</sup>“Society will not be the same after lockdowns are lifted,” *The Economist*, March 25, 2020.

## 2.2 Emergency Visits

The data for emergency visits comes from daily public reports by the Chilean Ministry of Health. The data includes the number of ED visits in each hospital in Chile, split by categories of related diagnoses.<sup>10</sup> In this paper we focus on non-respiratory visits, which accounted for 84 percent of the total visits in the period of analysis.<sup>11</sup> We show results for all non-respiratory visits combined, as well as for the main sub-categories available in the data: trauma or poisoning (that correspond to 16 percent of non-respiratory visits), circulatory system (3 percent), diarrhea (5 percent), and all other non-respiratory causes (76 percent).<sup>12</sup>

Figure 1 displays the total number of all non-respiratory ED visits in Chile for 2019 and 2020 at the daily level. Like in the U.S. and the U.K., there was a sharp drop in emergency visits in Chile after the beginning of the pandemic, in mid-March 2020. Non-respiratory visits declined by 52 percent in the post-March 13 period relative to the pre-March 13 period.<sup>13</sup>

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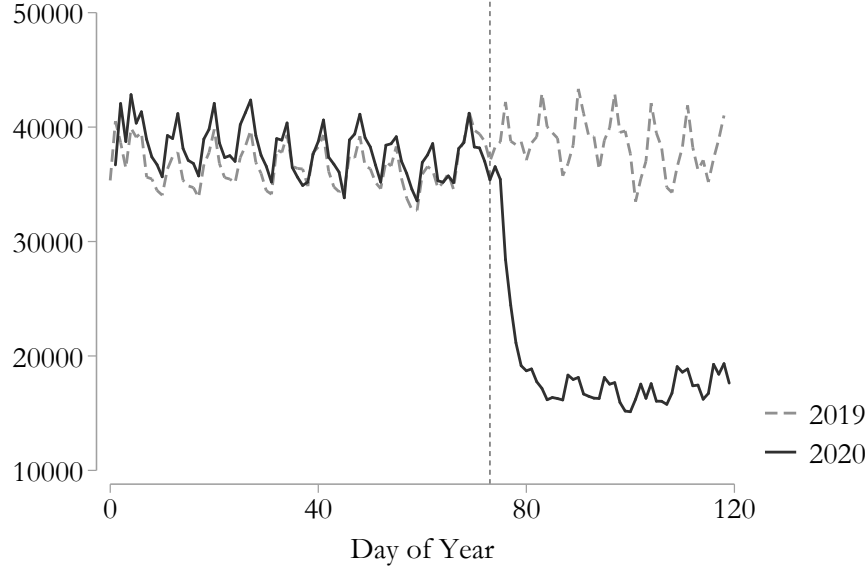
<sup>10</sup>The cases are classified using the ICD classification method. Each category in the data corresponds to a group of ICD codes.

<sup>11</sup>We exclude all respiratory causes from our analysis in order to eliminate potential visits related to COVID-19. These comprise acute bronchitis (J20-J21), influenza (J09-J11), pneumonia (J12-J18), and other respiratory causes (J22; J30-J39, J47, J60-J98).

<sup>12</sup>Although we do not have more disaggregated causes, the top (non-respiratory) diagnoses in the U.S. are chest pain, urinary tract infection, headache, unspecified abdominal pain, syncope and collapse, non-infective gastroenteritis and colitis, dizziness and giddiness, and low back pain (<https://blog.definitivehc.com/top-20-most-common-er-diagnoses>.)

<sup>13</sup>Figure ?? describes the change in ED visits for each particular cause identified in the data. A difference-in-differences analysis shows that non-respiratory visits decreased 57 percent relative to the pre-pandemic average when using 2019 as the control group, and 66, 38, and 92 percent for trauma or poisoning, circulatory system, and diarrhea visits, respectively. See Appendix A3 for more details.

**Figure 1:** Non-Respiratory Emergency Room Visits in Chile in 2019 and 2020



Note: The graph shows total emergency visits in Chile for the days after the first Wednesday of the year for the years 2019 and 2020 (January 2 and January 1, respectively) in all non-respiratory causes. The vertical line indicates March 13, 2020, which represents the last weekend before the drop in mobility and the implementation of the first set of social distances measures adopted by the government.

## 2.3 Descriptive Statistics

Table 1 shows the mean and the standard deviation of ED visits (Panel A) and of the mobility indices (Panel B) for the main two periods of analysis: before and after the beginning of the pandemic. Panel A summarizes the previous results shown in Figure 1 by averaging the daily number of ED visits by period on each category. Panel B also shows a “non-residential” index that we construct as the principal component of all out-of-residencies indices, standardized using its pre-pandemic mean and average.<sup>14</sup> The panel B in Table 1 shows a large change in all the variables. For example, relative to the pre-COVID-19 period, the number of visits and length of stay at workplaces during the COVID-19 period decline by 4.16 standard deviations. In turn, the residential index increases by 11 standard deviations.

<sup>14</sup>As we discuss in section 3, we use a principal component analysis due to the high collinearity in the mobility indices.

**Table 1:** Descriptive statistics by period of analysis

|                               | Before March 13       | Post March 13         |
|-------------------------------|-----------------------|-----------------------|
| Panel A: ER visits            |                       |                       |
| All Non-Respiratory           | 2,337.19<br>(2984.70) | 1,159.75<br>(1552.19) |
| Trauma or Poisoning           | 380.56<br>(499.07)    | 175.41<br>(244.14)    |
| Circulatory System            | 61.93<br>(84.50)      | 43.60<br>(60.33)      |
| Diarrhea                      | 144.91<br>(155.79)    | 42.76<br>(59.80)      |
| All Other Causes              | 1,749.78<br>(2264.47) | 897.98<br>(1203.75)   |
| Panel B: Mobility Index       |                       |                       |
| Residential                   | 0.72<br>(1.76)        | 20.40<br>(6.17)       |
| Non-Residential Summary (PCA) | 0.00<br>(1.00)        | -4.61<br>(1.43)       |
| Workplaces                    | 4.36<br>(10.45)       | -36.55<br>(16.68)     |
| Retail and Recreation         | 1.13<br>(8.53)        | -59.27<br>(16.86)     |
| Grocery and Pharmacy          | 3.67<br>(7.39)        | -37.34<br>(18.87)     |
| Parks                         | -5.99<br>(19.93)      | -61.04<br>(13.60)     |
| Transit Stations              | 0.74<br>(10.24)       | -57.62<br>(17.78)     |

Notes: Panel A shows average daily number of emergency daily visits by region in each category. Panel B shows average Google Community Mobility Report indices by region in the different categories. Non-residential summary (PCA) represents a summary index of all out-of-residencies indices as described in section 3. Both panels show summary statistics before and after March 13. Standard deviations are reported in parenthesis.

### 3 Estimating the Relationship between Mobility and ED Visits

The goal of this paper is to examine the degree to which the demand for ED visits can be explained by changes in mobility. In this section, we estimate a very simple model at the daily-regional level to predict ED visits. In the next section, we test the prediction capability of the our model by examining its ability to explain the abrupt drop in ED visits during the beginning of the COVID-19 pandemic.

#### 3.1 Empirical Strategy

We posit a simple linear model relating ED visits in day  $d$ , for diagnosis  $k$  and region  $r$  to mobility:

$$Y_{dkr} = f_{kr}(\mathbf{M}_{dr}) + \mu_{dow(d),k} + \epsilon_{dkr} \quad (1)$$

where  $f_{kr}$  is a type- and region-specific function,  $\mathbf{M}_{dr} \in R^6$  is a vector containing the daily measures of the Google Mobility Report in the six categories of places; and  $\mu_{dow(d),k}$  are type-specific dummies for the day of week (Monday, Tuesday, etc.), which absorb the intra-week variation of visits.<sup>15</sup>

In our first specification we use a parsimonious linear model for  $f_{kr}$ : Given the high collinearity of the mobility indices, instead of using the six the mobility indices directly, we summarize them with two indices. The first,  $Z_{dr}^{res}$ , is simply the normalized residential mobility index, and the second,  $Z_{dr}^{non-res}$ , is the first principal component derived from all five non-residential indices (also normalized).<sup>16</sup> For ease of interpretation, we standardize both indices at their pre-March 13 level. Thus, our first specification is  $f_{kr}^a = \alpha_{kr} + \beta_{k1}Z_{dr}^{res} + \beta_{k2}Z_{dr}^{non-res}$ . We estimate the coefficients of this first specification via OLS.

In our second specification we allow  $f_{kr}$  to contain all the original mobility indices provided by Google ( $M_{dr1}, M_{dr2}, \dots, M_{dr6}$ ), as well as a full set of interaction terms between the region and the mobility indices. The specification thus becomes:  $f_{kr}^b = \alpha_{kr} + \sum_{j=1}^6 \beta_{krj}M_{drj}$ . We estimate this model via lasso.<sup>17</sup> We view these two methodologies as complementary: On the one hand, lasso is a procedure designed to maximize the prediction of the model, and

<sup>15</sup>There is no intra-week seasonality in the mobility data due to the way it was constructed.

<sup>16</sup>The non-residential component explains 68 percent of the variance of the non-residential indices. In A5.2 we show that our main results are robust to different ways of performing the principal component analysis.

<sup>17</sup>We estimate lasso using the implementation of Friedman et al.'s (2010) coordinate descent algorithm. We use the code by Townsend (2018). Also, in reality we implement elastic net, which is a generalization of lasso, but the elastic-net penalty found by the algorithm is the same as lasso's.

therefore it is useful to estimate a flexible relationship between activity and ED visits, and allows us investigate the full extent to which the activity variables can explain the drop in ED. On the other hand, OLS provides results from a more parsimonious and easily-interpretable model, and allows us to do inference on the predictions.

As noted above, we estimate the model using the 2020 data for the *pre-COVID-19 period only*. Then, we compute the predicted values,  $\hat{Y}_{dkr}$ , for the entire sample: the pre-COVID-19 period (in-sample prediction); and post-COVID-19 period (out-of-sample prediction). Finally, we compute the country-level predicted totals as the sum of the regional predictions  $\hat{Y}_{dk} \equiv \sum_r \hat{Y}_{dkr}$ .

## 3.2 Results

Table 2 shows the estimation results of Equation (1) by OLS in the pre-pandemic period.<sup>18</sup> We find a significant relationship between the pre-COVID-19 emergencies and the pre-COVID-19 mobility indices. Overall, we find that a one standard deviation increase in the residential index decreases the average daily visits by 34 (or 1.5 percent of the sample average). Also, a one standard deviation increase in the non residential index increases ED visits increase the average daily visits by 52 (or 2.2 percent of the sample average).<sup>19</sup>

The  $F$ -statistic (and its corresponding  $p$ -value) for the joint test that both coefficients of the mobility variables equal zero in the pre-COVID-19 (training) sample ( $\beta_{k1} = \beta_{k2} = 0$ ) is  $F = 8.85(0.00)$  for the group of all non-respiratory visits. However, we do find heterogeneity in the explanatory power of the mobility variables across the different types of diseases. The mobility variables have the highest predictive power (as measured by the  $F$ -statistic) for trauma or poisoning visits ( $F = 6.05(0.00)$ ) and for the category including all other causes ( $F = 7.86(0.00)$ ). On the other hand, mobility indices have the lowest predictive power for visits related to Diarrhea ( $F = 2.40(0.09)$ ). These results are consistent with Evans and Moore (2012) who find that the relationship between economic activity and within-month patterns is stronger for trauma and substance-abuse deaths than for heart attacks and other heart-related deaths.

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<sup>18</sup>We use Newey and West (1987) standard errors to allow for autocorrelation for up to 3 days (following Greene, 2011). However, the standard errors of all the estimates in the paper almost do not change when using, for example, 7 daily lags.

<sup>19</sup>Given the day of the week fixed effects, the identification of the estimates does not come from intra-week patterns.

**Table 2:** Activity and ED visits in the pre-COVID-19 period

|                 | (1)<br>All<br>Non-Respiratory | (2)<br>Trauma or<br>Poisoning | (3)<br>Circulatory<br>System | (4)<br>Diarrhea  | (5)<br>All Other<br>Causes |
|-----------------|-------------------------------|-------------------------------|------------------------------|------------------|----------------------------|
| Residential     | -33.98***<br>(13.89)          | -12.95***<br>(4.36)           | 0.11<br>(0.60)               | -1.21<br>(1.38)  | -19.94**<br>(11.19)        |
| Non Residential | 52.18***<br>(14.49)           | 9.26***<br>(3.11)             | 1.31**<br>(0.60)             | 2.71**<br>(1.86) | 38.90***<br>(11.87)        |
| N               | 448                           | 448                           | 448                          | 448              | 448                        |
| $R^2$           | 0.994                         | 0.986                         | 0.979                        | 0.968            | 0.992                      |
| F-stat.         | 8.85                          | 6.05                          | 2.57                         | 1.34             | 7.86                       |
| p-value         | 0.00                          | 0.00                          | 0.08                         | 0.26             | 0.00                       |
| Mean dep. var.  | 2337                          | 381                           | 62                           | 145              | 1750                       |

Notes: The table shows the results of a OLS estimation of Equation (1) in the pre-COVID-19 period. Robust standard errors possibly autocorrelated up to 3 lags (Newey and West, 1987). \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

### 3.3 Robustness

In Table A2 of the Appendix we show that the relationship between ED visits and mobility is robust to a series of different specifications. First, excluding the day-of-the-week fixed effects, the relationship between mobility and ED visits is only slightly weaker.<sup>20</sup> Second, the results are robust to eliminating March 8 from the sample, a day when ED visits were relatively high and there was also a spike in the residential mobility index due to demonstrations as part of the International Women’s day. Third, the relationship is robust to adding month fixed-effects, and to adding month fixed-effects as well as their interaction with the day-of-the-week fixed effects. Overall, these robustness exercises show that it is the high-frequency variation that used to estimate the relationship between visits and mobility instead of potential seasonal patterns. See Appendix A5.1 for more details.

<sup>20</sup>Although this result may seem counterintuitive, it is important to keep in mind that the normalization of the mobility variables is specific to the day of the week, and therefore part of the variation across days of the week is eliminated.

## 4 Predicting ED Visits with Mobility Data: A case study

In the previous section, we estimated the relationship between mobility (as a proxy for social and economic activities and risk exposure) and the various types of ED visits using data before the COVID-19 crisis. In this section, we use those estimates to construct the emergency visits predicted by the model under the post-COVID-19 mobility patterns. We use this prediction exercise to quantify the share of the decrease in visits that can be attributed to the changes in mobility.

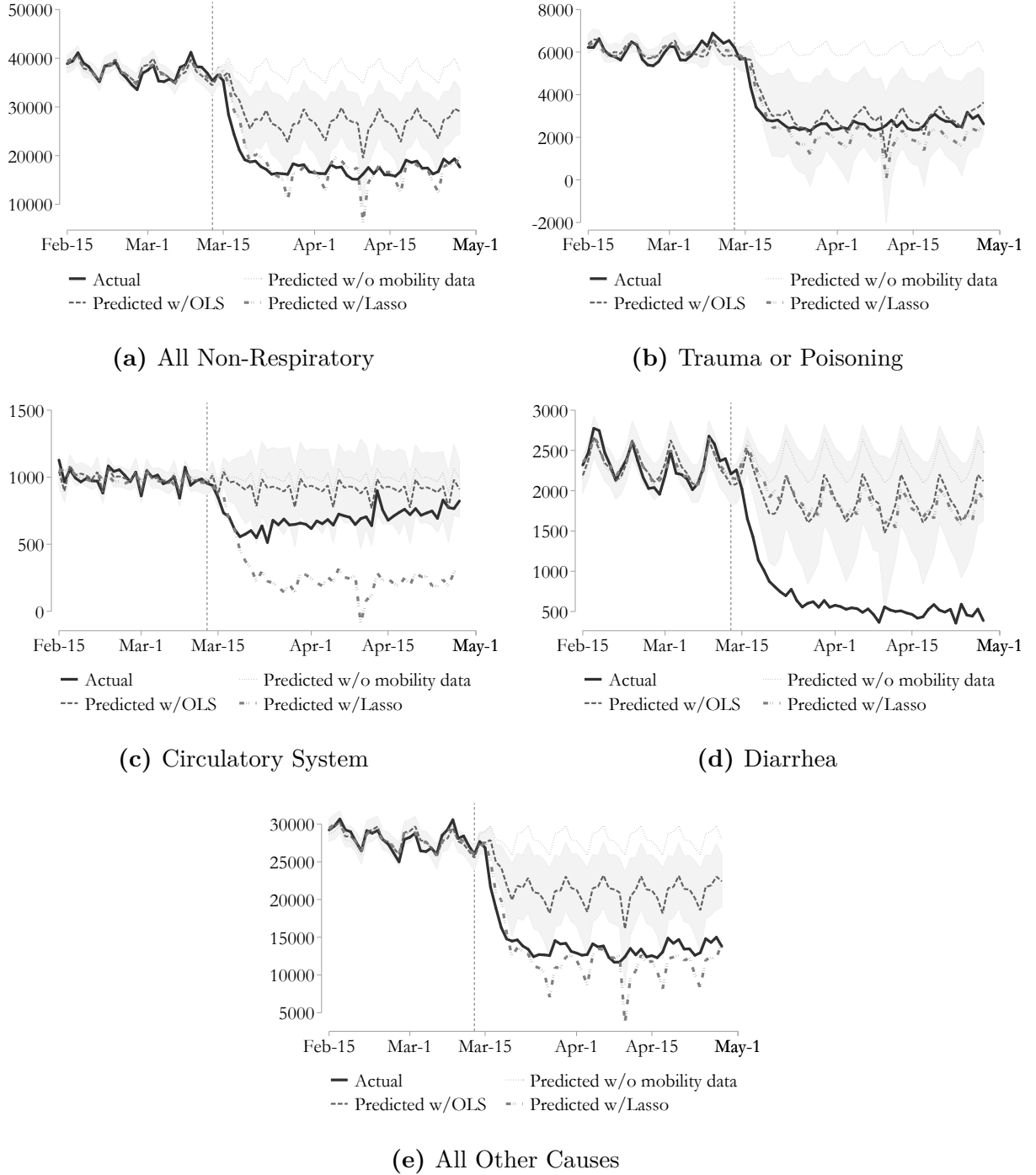
### 4.1 Visually Comparing Predicted and Actual ED visits using *out-of-sample* Data

We start with a visual depiction of the results. Figure 2 contrasts the evolution of ED visits to the predictions of several models. Specifically, it shows (i) the actual time series, (ii) the prediction from a model that only includes day-of-the-week and region fixed effects, and omits the mobility data; (iii) the OLS prediction and its confidence intervals, and (iv) the lasso prediction. Panel (a) shows the results for all (non-respiratory) visits, while panels (b) to (d) correspond to each type.

We find that the mobility patterns explain a large fraction of the decrease in the emergency visits. Overall, all non-respiratory visits dropped from an average of 37,959 per day to an average of 18,555 visits per day, a 50.4 percent decrease after the onset of the crisis. As shown in Panel (a) of Figure 2, our OLS model predicts roughly half of that drop. Moreover, the lasso prediction for the drop is almost equal to the actual drop.

Panels (b)-(e) of Figure 2 show that the ability of our model to explain the drop in ED visits is heterogeneous across types. Our model predicts a fair share of trauma or poisoning, and other non-respiratory visits, but is not able to consistently explain the drop in visits related to the Circulatory System or Diarrhea. These facts are consistent with the results in Table 2) that show that the mobility indices are statistically correlated with trauma or poisoning, and other non-respiratory, but not statistically correlated with Circulatory System or Diarrhea ED visits in the pre-COVID period.

**Figure 2: Actual and Predicted Emergencies by Type**

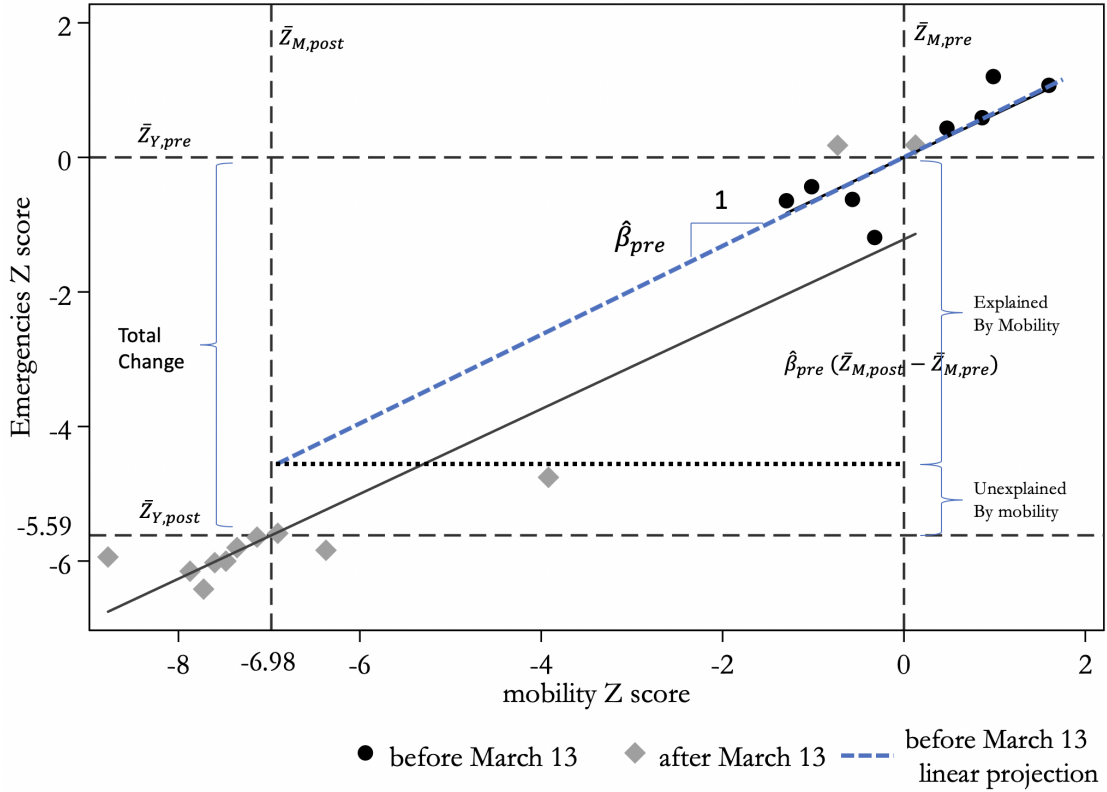


Note: The panels show the predicted and the actual times series of the ED visits in the different categories. See the text for the construction of each of the predicted lines. The grey area indicates the OLS confidence intervals of the model with mobility data. The vertical line indicates the end of the prediction period (March 13, 2020).

## 4.2 Quantifying the Role of Mobility

We now formally quantify the ability of our model to predict the abrupt changes in ED visits experienced during 2020 at the beginning of the pandemic. Figure 3 illustrates the basic intuition of our approach. The figure plots the Z-score of the index for mobility to transit stations against the Z-score of the trauma or poisoning visits in the capital city (Santiago), and the linear fits for each the pre-pandemic and for the pandemic periods. Z-scores for the entire sample period are based on the means and standard deviations in the pre-COVID period. Therefore, the magnitude in both axes corresponds to the deviation of a particular measurement with respect to its pre-COVID mean as fraction of its pre-COVID standard deviation.

**Figure 3:** Empirical Approach



Note: The figure shows a binned scatter plot of the normalized (Z-score) transit stations mobility and normalized (Z-score) trauma and poisoning emergency visits in the Metropolitan region (Santiago and its surroundings). The normalizations use the corresponding pre-March 13 mean and variances in Santiago. The straight lines represent the linear fit for each period.

The linear fit for the data before March 13 indicates that, in days when the mobility index is one standard deviation lower than average, there are also (on average) 0.64 standard deviations less ED visits due to trauma or poisoning. In the case of a simple univariate

model for ED visits, our empirical strategy would consist of (i) estimating the slope  $\hat{\beta}_{pre}$  using the pre-COVID period, and (ii) using  $\hat{\beta}_{pre}$  to project the linear relationship up to the mean mobility in the post-COVID period  $\bar{Z}_{M,post}$ . The share of total change in average visits  $\bar{Z}_{Y,post} - \bar{Z}_{Y,pre}$  that is explained by the change in mobility corresponds to  $\hat{\beta}_{pre}(\bar{Z}_{M,post} - \bar{Z}_{M,pre})$ . In the example presented in Figure 3, the transit stations mobility index alone would allow us to explain 80% of the decline in ED visits for Santiago.<sup>21</sup> Our empirical exercise takes uses the insight described in Figure 3, but allows for a relationship between ED visits and the six different mobility indices of the Google Mobility Reports, while controlling for various time and region fixed effects.

Our quantifying exercise is akin to the (three-fold) Kitagawa-Oaxaca-Blinder decomposition (KOB) of gender wage gaps used in the labor economics literature (Kitagawa, 1955; Blinder, 1973; Oaxaca, 1973). This method is traditionally used to study the wage gap across groups (e.g., males vs. females) by decomposing the gap into the part that can be explained by observable characteristics (e.g., differences in “endowments,” such as education in the gender wage gap literature) and the part of the gap that cannot be explained by observables. We apply the same logic to decompose differences in average visits across the two groups of observations defined by the calendar time: the group of post-March 13 days and the group of pre-March 13 days. The goal of the decomposition is to quantify the part of the difference in average visits across the two groups that can be explained by the model, particularly by the mobility variables. We note that the decomposition does *not* hinge on the assumption that the model relating emergencies and activity remains the same after the pandemic. In fact, the decomposition exercise is precisely meant to split the change in ED visits into the part that can be explained by the pre-pandemic model (the difference attributed to “endowments” in the context of a KOB decomposition) and the part that is due to a change in the model (the “unexplained” component).

To simplify the notation, we rewrite the general model for average visits in each period as:

$$\begin{aligned} E[Y_{pre}] &= \bar{X}'_{pre} \beta_{pre} \\ E[Y_{post}] &= \bar{X}'_{post} \beta_{post} \end{aligned}$$

A “three-fold” decomposition of the gap  $E[Y_{pre}] - E[Y_{post}]$  can be written as:

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<sup>21</sup>The average transit stations mobility score in the pandemic is  $\bar{Z}_{M,post} = -6.98$ . The linear projection of the pre-March-13 relationship predicts a drop of  $0.64 \times 6.98 = 4.48$ , which corresponds to 80% of the actual decline ( $\bar{Z}_{Y,post} = 5.59$ ).

$$E[Y_{post}] - E[Y_{pre}] = (\bar{X}_{post} - \bar{X}_{pre})'\beta_{pre} + \bar{X}'_{pre}(\beta_{post} - \beta_{pre}) + (\bar{X}_{post} - \bar{X}_{pre})'(\beta_{post} - \beta_{pre}) \quad (2)$$

The first part of this decomposition,  $(\bar{X}_{post} - \bar{X}_{pre})'\beta_{pre}$ , corresponds to the difference in average visits across periods that can be explained by observables. This component corresponds to the part of the gap that can be explained by extrapolating the pre-COVID-19 relationship ( $\beta_{pre}$ ) onto the post-COVID-19 average mobility ( $\bar{X}_{post}$ ) (see Figure 3). Consequently, the sum of the second term and the third term  $\bar{X}'_{pre}(\beta_{post} - \beta_{pre}) + (\bar{X}_{post} - \bar{X}_{pre})'(\beta_{post} - \beta_{pre})$  corresponds to the part of the gap that cannot be explain by observables.

Dividing each side of Equation (2) by  $E[Y_{pre}]$  we can re-write the equation as

$$\underbrace{\frac{E[Y_{post}] - E[Y_{pre}]}{E[Y_{pre}]}}_{\text{Difference \%}} = \underbrace{\frac{(\bar{X}_{post} - \bar{X}_{pre})'\beta_{pre}}{E[Y_{pre}]}}_{\text{Explained by Observables \%}} + \underbrace{\frac{\bar{X}'_{pre}(\beta_{post} - \beta_{pre}) + (\bar{X}_{post} - \bar{X}_{pre})'(\beta_{post} - \beta_{pre})}{E[Y_{pre}]}}_{\text{Unexplained by Observables \%}}, \quad (3)$$

which results in the share of the explained and unexplained gap in percentage terms.

Table 3 shows the result of this decomposition, where each panel corresponds to a different type of ED visit. The last row of each panel presents the standard error of the OLS explained share in parenthesis. Panel (A) shows that all non-respiratory visits dropped from 37,959 to 18,555 visits per day, a 50.4 percent decrease after the onset of the crisis. Our OLS model predicts a drop of 26.3 percent. Thus, a simple linear model of the Google mobility indices predict 52.1 percent of the total drop in non-respiratory ED visits. Moreover, the lasso regression is able to explain 100.4 percent of the overall drop.

The other panels of Table 3 repeat the decomposition for other causes. Our model explains 91 percent (29.7 percent s.e.) of the drop of Trauma and Poisoning visit, and 47 percent (17.4 percent s.e.) for the group labeled as Other Causes, which jointly make up for more than 90 percent of the ED visits in our estimation sample. Also, Panels (C) and (D) shows the results for circulatory system and diarrhea visits. The panels indicate that the standard errors of these OLS predictions are large, fact which was expected given the low  $F$  statistics of the regression of these visits on the mobility indices.

### 4.3 Robustness

The fraction of the drop in ED visits that is explained by our model is robust to a number of different ways of performing the principal component analysis. Specifically, Appendix A5.2 shows that the results are robust to the following ways of performing the principal

**Table 3:** Decomposition of the Drop in Emergency Visits

|                                   | (1)                 | (2)       | (3)    |
|-----------------------------------|---------------------|-----------|--------|
|                                   | Observed            | Predicted |        |
|                                   |                     | OLS       | Lasso  |
| Panel A:                          | All Non-Respiratory |           |        |
| Before March 13 (A1)              | 37,395              | 37,395    | 37,395 |
| After March 13 (A2)               | 18,555              | 27,571    | 18,482 |
| Difference (%) ( $[A2 - A1]/A1$ ) | -50.4%              | -26.3%    | -50.6% |
| Difference Explained by Model (%) |                     | 52.1%     | 100.4% |
|                                   |                     | (16.61)   |        |
| Panel B:                          | Trauma or Poisoning |           |        |
| Before March 13 (A1)              | 6,090               | 6,090     | 6,089  |
| After March 13 (A2)               | 2,806               | 3,110     | 2,324  |
| Difference (%) ( $[A2 - A1]/A1$ ) | -53.9%              | -48.9%    | -61.8% |
| Difference Explained by Model (%) |                     | 90.7%     | 114.7% |
|                                   |                     | (29.74)   |        |
| Panel C:                          | Circulatory System  |           |        |
| Before March 13 (A1)              | 991                 | 991       | 991    |
| After March 13 (A2)               | 698                 | 915       | 293    |
| Difference (%) ( $[A2 - A1]/A1$ ) | -29.6%              | -7.7%     | -70.5% |
| Difference Explained by Model (%) |                     | 26.0%     | 238.0% |
|                                   |                     | (47.12)   |        |
| Panel D:                          | Diarrhea            |           |        |
| Before March 13 (A1)              | 2,318               | 2,318     | 2,319  |
| After March 13 (A2)               | 684                 | 1,910     | 1,885  |
| Difference (%) ( $[A2 - A1]/A1$ ) | -70.5%              | -17.6%    | -18.7% |
| Difference Explained by Model (%) |                     | 25.0%     | 26.5%  |
|                                   |                     | (20.54)   |        |
| Panel E:                          | All Other Causes    |           |        |
| Before March 13 (A1)              | 27,997              | 27,997    | 27,996 |
| After March 13 (A2)               | 14,368              | 21,637    | 12,850 |
| Difference (%) ( $[A2 - A1]/A1$ ) | -48.7%              | -22.7%    | -54.1% |
| Difference Explained by Model (%) |                     | 46.7%     | 111.1% |
|                                   |                     | (17.39)   |        |

Note: The table shows the KOB decomposition for the observed drop in ED visits. We report the share of the drop that can be explained by OLS and Lasso models. Column (2) includes robust standard errors possibly autocorrelated up to 3 lags (Newey and West, 1987) in percentage points of the share explained by the model.

component analysis: (1) replacing the residential and non-residential mobility indices with the two principal components that explain the most variance of the six mobility indices, (2) using three principal components, and (3) leaving out the index for “parks” and using two principal components for the other four non-residential indices.<sup>22</sup>

## 5 Additional Evidence

This section provides additional evidence on the relationship between social and economic activity and ED visits. We first examine intraweek patterns in ED visits. Second, we show evidence related to mortality data.

### 5.1 Intraweek Patterns

We take advantage of our daily data to examine the intraweek patterns of visits, and how these patterns changed after the pandemic. We show that: *(i)* different intraweek patterns between summer and fall of ED visits in the year before the pandemic show that ED visits are related to social and economic activity, and *(ii)* changes in the intraweek patterns during the pandemic are consistent with the idea that the change in social and economic activity played a role in the drop in ED visits. This evidence provides an explanation for why our model, which is estimated using data from mid-February to mid-March (the summer season in Chile), fits the (out-of-sample) pandemic period well.

Figure 4 plots the percent difference in ED visits for each day of the week relative to Sunday by visit category, and their corresponding 95 percent confidence interval. The figure shows this difference for the pre-pandemic and the pandemic periods in 2020, and for the same corresponding dates in 2019.<sup>23</sup>

Figure 4 shows two main findings. First, the figure shows that for non-respiratory visits, and particularly for the visits related to trauma and poisoning, the patterns change significantly between the corresponding “pre-COVID-19” period of 2019 and “COVID-19” period of 2019 (Panels (a) and (b) of Figure 4). The pre-COVID-19 periods contains mostly summer months, when schools are closed and many people are in vacation. These patterns from the year before the COVID pandemic strongly suggests that ED visits for these categories are linked to social and economic activity. Visits related to the circulatory system and to diarrhea show similar intraweek patterns in 2019. This evidence provides further evidence of

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<sup>22</sup>The article by “The Economist” cited above also constructs its economic activity index without the index for parks.

<sup>23</sup>In 2020, we use dates before March 13 and after March 26 period because we drop the transition period. For 2019 we use dates before March 15 and after March 26.

a weak link between social and economic activity for these causes, as discussed in subsection 3.2.

The second finding relates to patterns during the pandemic: Figure 4 shows that the pre-COVID-19 period in 2020 matches the corresponding period in 2019 well. In contrast, the COVID-19 period in 2020 follows the “summer patterns” of all non-respiratory causes and of trauma or poisoning much more closely than the patterns of the corresponding period of 2019. This is suggestive evidence that during the pandemic the relationship between economic activity and ED visits in those causes is similar to the relationship in the summer months, but not to the relationship in the corresponding period of 2019. This, in turn, provides a reason for why estimating the mobility-ED visits relationship during the summer months (as we did in Section 3.1) allows us to fit the pandemic period data well.<sup>24</sup>

## 5.2 Evidence from Mortality Data

Evidence from death records to show that the decrease in ED visits was not accompanied with a significant increase in mortality. The absence of a surge in mortality provides evidence that individuals with life-threatening conditions did not stop from going to the ED due to the fear of contracting the virus at the hospital.

We use death records for all reported deaths at the region-week level from January 1, 2016 to April 28, 2020.<sup>25</sup> Figure 5 shows the number of deaths in 2020 as well as the average in previous years. On average, there were roughly 2,000 deaths per week in 2020. Despite the higher level of deaths in 2020 with respect to previous years, we do not find discernible changes for the period after March 13, when emergencies decreased. We formalize the graphical analysis by estimating a difference-in-differences model. The results, which we show in Column (1) of Table 4, show that there was not a significant increase in deaths after April 28 concurrently with the decrease in ED visits.

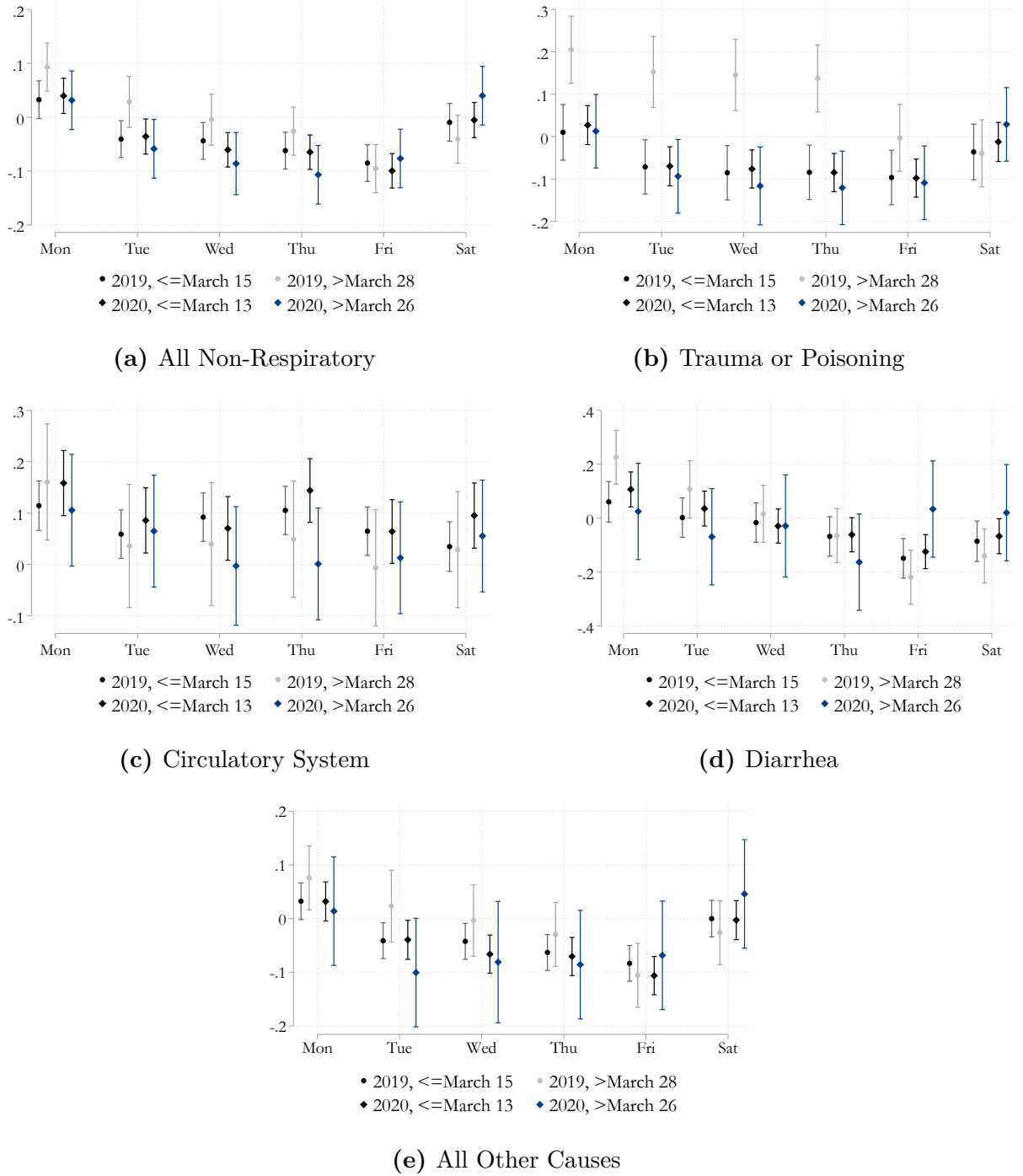
Why were there not fewer deaths? If most of the drop in ED was due to a lower incidence of conditions, in principle we may have expected a drop in deaths in the COVID-19 period. There are two possible explanations for why overall deaths remained stable despite the large drop in the number of ED visits. First, the stability in the overall number of deaths might mask a decrease in mortality due to lower mobility (e.g., due to fewer car accidents) that is coupled with an offsetting increase in the mortality rate due to COVID-19 or due to other

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<sup>24</sup>The model is estimated within a range of mobility values that is very different from the one during the pandemic. The fact that we fit the data during the pandemic also suggests that the linear model is a good specification in the entire range of observed mobility.

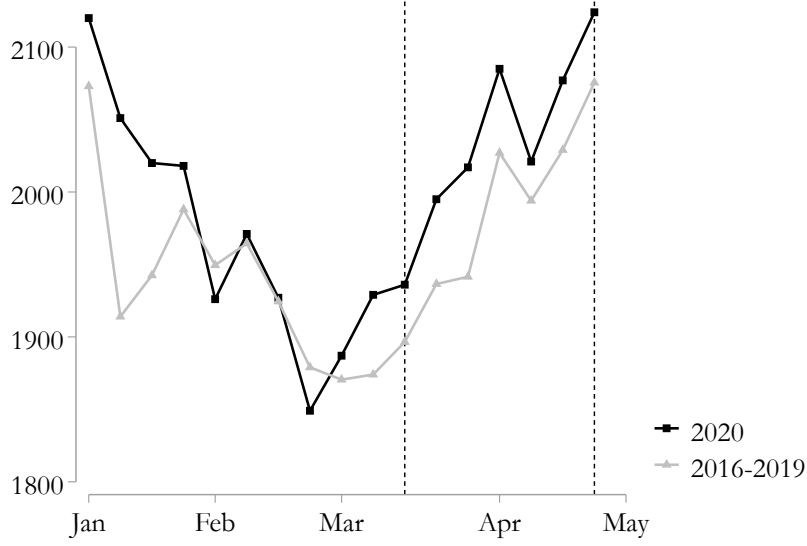
<sup>25</sup>The data includes causes of death causes, and was released by the Department of Health Statistics and information (DEIS) of the Ministry of Health (*Servicio de Registro Civil*), the governmental agency which manages official birth and death records, and is available at <https://deis.minsal.cl/>.

**Figure 4: Excess Visits relative to Sundays**



Note: The figure plots OLS estimates of the log number of visits on day-of-the-week indicators, using Sunday as the omitted category. For 2020, we split the sample into two groups: the pre-March 13 period and the post-March 26, after the post-COVID-19 visits stopped decreasing (see Figure ??). For 2019 we use the equivalent dates (pre-March 15 and post-March 26).

**Figure 5:** Number of deaths per week: 2010-2020



Note: The plot shows the number of weekly deaths over time. The solid black line and the gray line show the evolution of weekly deaths in 2020 and the 2016-2019 period, respectively. The vertical line indicates the beginning of the COVID-19 period.

conditions due to the fear of visiting the ED and contracting the virus at the hospital. We call this the “offsetting deaths” hypothesis. An alternative explanation is that the lower incidence of emergencies was mostly concentrated among cases with low mortality. Under this “selection” hypothesis we would expect lower ED visits without a corresponding decrease in mortality.

The data supports the selection hypothesis. We present evidence from two different analyses. First, we study cause-specific deaths data.<sup>26</sup> Table 4 shows difference-in-differences estimates for cause-specific changes in mortality. We estimate a differences-in-differences model for the number of deaths for different subset of causes:

$$\log(1+\text{Deaths})_{rwy} = \gamma_1 \text{Post}_w + \gamma_2 \text{Post}_w \times 1\{y = 2020\} + \psi_r + \phi_y + \epsilon_{rwy}. \quad (4)$$

The dependent variable is the log number of (one plus) deaths reported in a given region-week-year,  $\text{Post}_w$  refers to the period after week 11 (around March 13) for every year, and  $\psi_r$  and  $\phi_y$  are region and year fixed effects, respectively. Our coefficient of interest is  $\gamma_2$ , which measures the extra number of deaths reported during the COVID-19 period relative

<sup>26</sup>Separating COVID-19 deaths from other causes of death is inherently challenging and was not without controversy in Chile. Although we use official statistics, we take the results below with some caution.

**Table 4:** Change in deaths during post-COVID-19 period

|                           | (1)<br>All       | (2)<br>Non-resp.<br>Incl. Covid | (3)<br>Non-Resp.<br>Excl. Covid | (4)<br>Trauma or<br>Poisoning | (5)<br>Circulatory | (6)<br>All Other<br>Causes |
|---------------------------|------------------|---------------------------------|---------------------------------|-------------------------------|--------------------|----------------------------|
| Post $\times$ (Year=2020) | 0.019<br>(0.024) | 0.029<br>(0.024)                | -0.014<br>(0.024)               | -0.164***<br>(0.053)          | 0.025<br>(0.040)   | -0.010<br>(0.026)          |
| N                         | 1440             | 1440                            | 1440                            | 1440                          | 1440               | 1440                       |
| $R^2$                     | 0.976            | 0.972                           | 0.973                           | 0.797                         | 0.915              | 0.963                      |
| Deaths in 2020 (%)        | 100              | 89                              | 87                              | 8                             | 24                 | 56                         |

Note: Coefficients represent the change in dependent variable after week 11 (March, 13 in 2020) in year 2020, relative to the same period of the year during 2016-2019. Sample includes all 16 Chilean regions and deaths between 2016-2020. Newey-West standard errors in parenthesis. Includes region-year fixed effects. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

to the deaths reported in the same period of time during the previous years, after controlling for year-specific differences in mortality with the year fixed-effects. We estimate the model for the first 18 weeks of each year to match the data of our main analysis.

Table 4 presents the estimates of Equation (4). As we reported above, Column (1) shows a non-significant increase in deaths during the COVID-19 period, which means that the drop in emergencies did not directly correspond with a statistically significant change in deaths. This suggests that the drop in emergencies was unlikely to be driven by a decline in life-threatening ED visits. Columns (2)-(5) investigate cause-specific mortality. After excluding respiratory and COVID related deaths, we find a 1.4 percent drop (standard error of 2.4 percent) in mortality. This decrease is the combination of a 16.4 (s.e.=5.3) percent drop related to Trauma or Poisoning causes, which make up 8 percent of all 2020 deaths during our study period; a 2.5 (s.e.=4) percent increase in Circulatory System deaths, which make up 24 percent of deaths; and a 1 (s.e.=2.6) percent drop related to All Other Causes, which make up 56 percent of total deaths. Thus, the evidence shows an absence of an increase in deaths that could have been due to life-threatening conditions not being properly cared for in EDs. Overall, we find no significant changes in mortality except for a large drop (16 percent) for Trauma or Poisoning.<sup>27</sup> These findings provide evidence against the “offsetting-deaths” hypothesis.

The second set of results that support the selection hypothesis comes from relative decreases in ED visits: Using 2019 as a control group, a differences-in-differences analysis reveals that the size of the decrease in ED visits was heterogeneous across types, and that the decrease was the largest for causes that are less fatal on average (Trauma and Poison-

<sup>27</sup>The results are qualitatively similar if instead of log-linear models we estimate Poisson models with wild bootstrap clustering at the regional level to correct for the small number of clusters.

**Table 5:** Change in Share of Visits and Mortality

| Type                      | ED Visit Share<br>Change (%)<br>(1) | Deaths-to-ED<br>Visits Ratio<br>(2) |
|---------------------------|-------------------------------------|-------------------------------------|
| Trauma or Poisoning       | -1.87***<br>(0.23)                  | 0.47%                               |
| Circulatory System        | 1.23***<br>(0.07)                   | 8.48%                               |
| Diarrhea and Other Causes | 0.64***<br>(0.24)                   | 0.63%                               |

Note: Column (1) presents differences-in-differences estimates for the change in the share of non-respiratory visits for the main ED visits causes between January 1 and April 28, where we use 2019 as the control group for 2020. The estimation includes day of the week, month of the year, year, and regional fixed effects. Robust and possibly autocorrelated (3 lags) standard errors (Newey and West, 1987) in parentheses. Column (2) shows the average weekly ratio between deaths and total non-respiratory ED visits for the period 2016-2019. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

ing), and the lowest for visits that are more fatal on average (Circulatory System).<sup>28</sup> Column (1) of Table 5 presents differences-in-differences estimates for the change in non-respiratory visits for each cause using a similar specification as in Equation (4). Column (2) shows the ratio between deaths and total non-respiratory ED visits, using death records for each type for the period 2016-2019.

Overall, Table 5 indicates that the category that decreased its share the most during the pandemic (Trauma or Poisoning) is also the least fatal category, with a 0.0042 death-to-visits ratio. On the other hand, the category that gained the most share (diseases of the circulatory system) is also the most fatal one, with a death-to-visits ratio of 0.084.

These results suggest that the drop in visits was *selective*, in the sense that the drop in ED visit was concentrated among the least fatal categories. Given that there was selection across categories, it is plausible that there was selection within categories as well, so that the drop in visits was concentrated in the less fatal cases even within categories. Sufficient selection could explain a large drop in ED visits happening concurrently with no significant drop in mortality.

<sup>28</sup>The fatality rate of each type of ED visit is computed as the ratio between the total ED visits and the number of deaths for each corresponding cause in the period 2016-2019.

## 6 Conclusion and Managerial Implications

We use novel daily data to demonstrate that social and economic activity, as proxied by mobility, affects demand for ED visits. We use the COVID-19 pandemic as a case-study to illustrate the predictive power of our model as a result of the changes in mobility during the pandemic. Specifically, we estimate a model of the effect of mobility on ED visits using data prior to the COVID-19 pandemic. We then apply the (out-of-sample) model predictions to show that the observed changes in population’s mobility can explain most of the drop in a particularly disruptive scenario. This demonstrates the out-of-sample performance of our model, even well outside the support of the data used to estimate it.

Our findings reveal a strong link between mobility and ED visits related to trauma, but a weaker link for visits less related to mobility, such as those for circulatory system issues or diarrhea. In the case of trauma, our model accurately predicts 100% of the observed drop in ED visits during our period of analysis.

Understanding the relationship between mobility and demand for ED services can help government administrators to accommodate services accordingly. Our findings suggest that incorporating city-wide or higher-level information on mobility is crucial for organizing the supply of ED services. These data can be particularly useful when anticipating larger changes due to school closures, holidays, vacations, or temporary closures of specific activities. Another important implication for public health is that policies that reduce human mobility to contain disease spread can also free up resources and benefit public health in the short term. Information about human mobility can be crucial for optimizing hospitals’ resource allocation and staffing decisions.

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# Online Appendix

## A1 The COVID-19 crisis in Chile

The confirmation of the first case of COVID-19 in Chile occurred on March 3, 2020. On March 15 the government announced the first set of measures restricting mobility. All these measures started on March 16, and mainly contemplated a nationwide closing of schools and educational establishments, restaurants, and movie theaters, as well as forbidding events with more than 50 people. Further measures adopted were a national night curfew between 10 PM and 5 AM starting on March 22, and localized lockdowns in specific municipalities starting on March 26.<sup>29</sup> On March 23, when there were 93 total positive cases, the first death related to COVID-19 was identified.

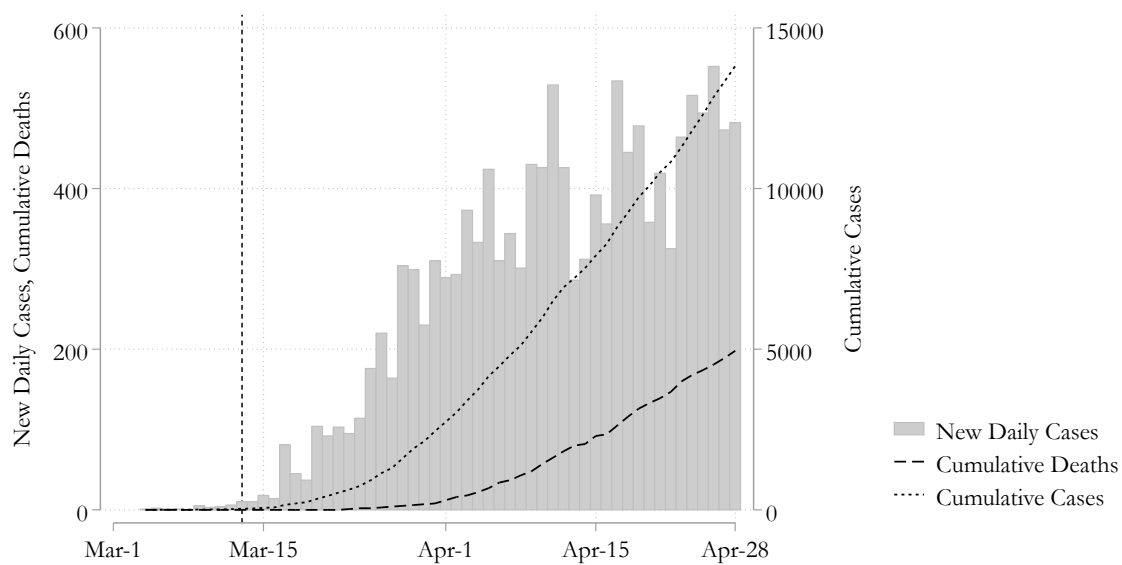
Figure A1 plots the number of new daily cases, the number of cumulative cases, and the number of deaths of COVID-19 in Chile.

Figure A2 shows evidence of no capacity constraints during the period of analysis. The figure shows that the increase in use of ICU beds by COVID-19 patients and the number of mechanical ventilators in use happened only after April 28, the date where our analysis ends, which is marked by the dashed line. The figure also shows that the total capacity of mechanical ventilators stayed almost constant before April 28.

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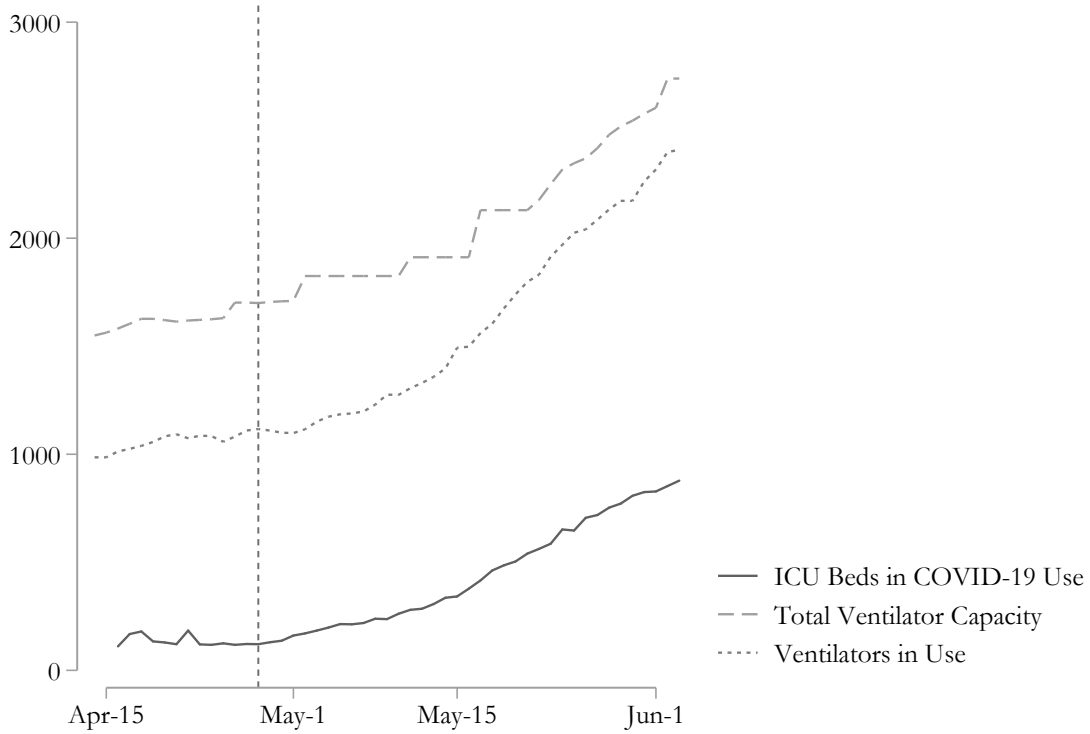
<sup>29</sup>Lockdowns were implemented selectively. Due to their initial higher contagion rate seven municipalities of the Santiago metropolitan area were under a total lockdown that lasted fifteen days. People under lockdown were required to obtain a special permit to move from one location to another. Later, thirteen additional municipalities across the country were under temporary lockdown at some point between March 30 and April 23.

**Figure A1:** Number of COVID-19 cases in Chile



Note: The graph shows the number of new daily cases, the number of cumulative cases, and the number of deaths of COVID-19 in Chile. The dashed vertical line indicates March 13. Source: Our World in Data <https://ourworldindata.org/coronavirus>.

**Figure A2:** ICU Beds in Use, and Ventilator Use and Capacity



Note: The figure shows the use of ICU beds by COVID-19 patients, the number of mechanical ventilators in use, and the total number of mechanical ventilators (in use or not). The dashed line indicates April 28, the date when our analysis ends. Source: Ministry of Science and Technology.

## A2 Google Mobility Report

The Google Mobility Report shows how visits and length of stay at different places change compared to a baseline. The baseline is the median value, for the corresponding day of the week, during the 5-week period Jan 3–Feb 6, 2020. The data was publicly available as of April 2020 in <https://www.google.com/covid19/mobility/>.

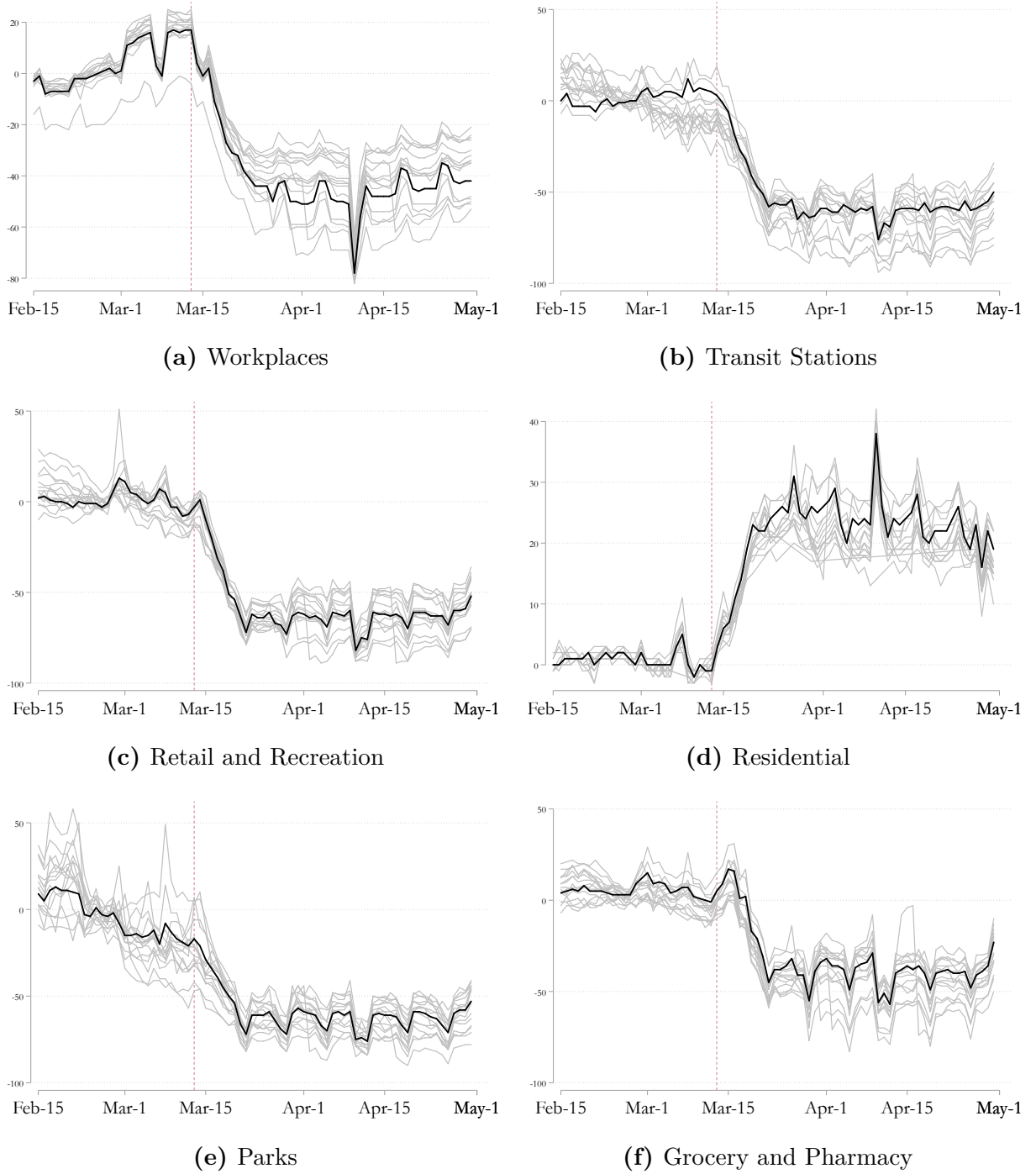
These changes are calculated using the same kind of aggregated and anonymized data used to show popular times for places in Google Maps. Mobility trends are split in the following categories:

1. Retail and recreation: Mobility trends for places like restaurants, cafes, shopping centers, theme parks, museums, libraries, and movie theaters.
2. Grocery and pharmacy: Mobility trends for places like grocery markets, food warehouses, farmers markets, specialty food shops, drug stores, and pharmacies.

3. Parks: Mobility trends for places like national parks, public beaches, marinas, dog parks, plazas, and public gardens.
4. Transit Stations: Mobility trends for places like public transport hubs such as subway, bus, and train stations.
5. Work: Mobility trends for places of work.
6. Residential: Mobility trends for places of residence.

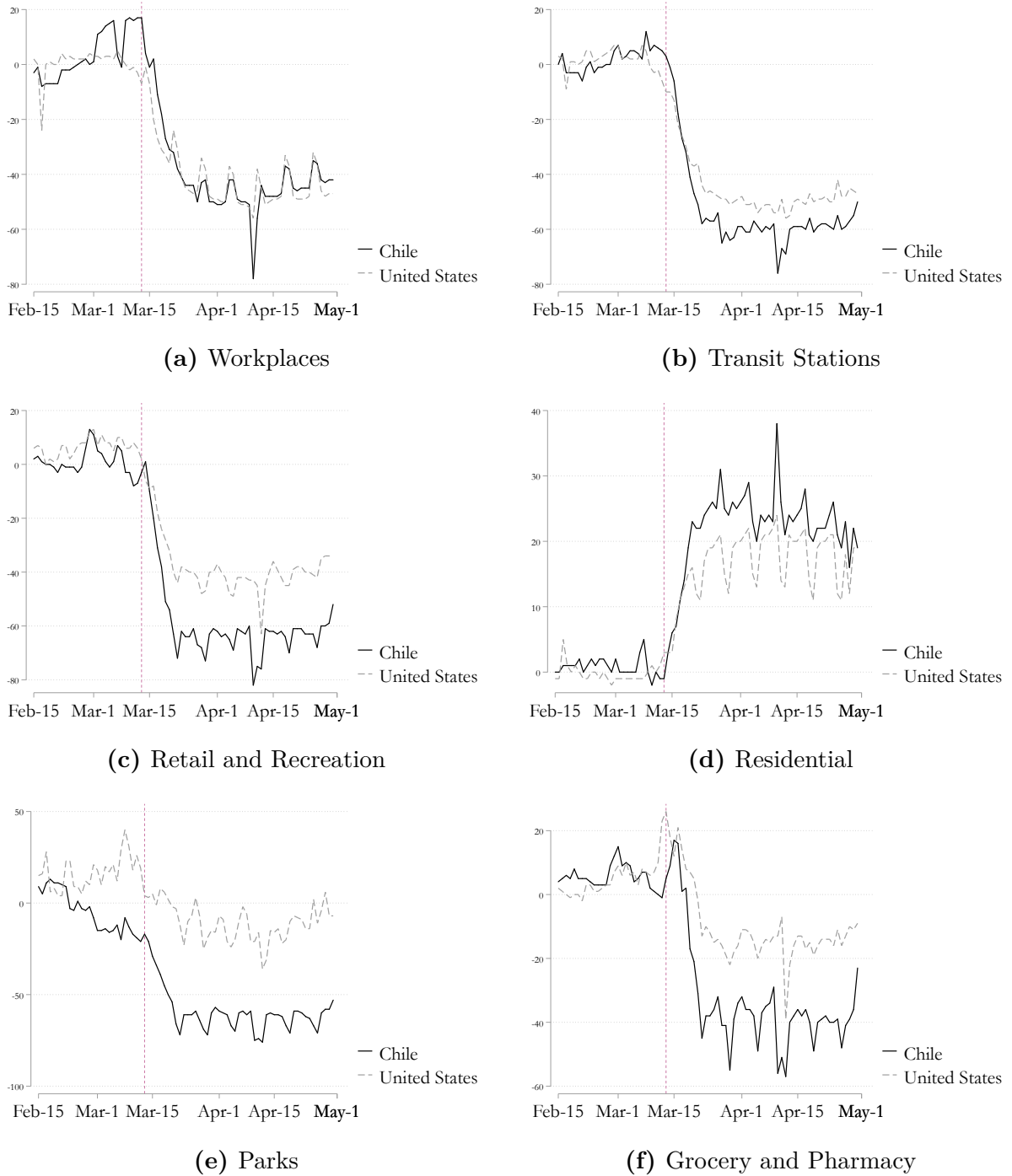
Figure A3 presents the mobility indices in Chile at the regional and national levels in grey and black lines, respectively. Figure A4 shows the mobility indices for Chile and the U.S. More details on how mobility index are calculated can be found in [https://www.google.com/covid19/mobility/data\\_documentation.html?hl=en#about-this-data](https://www.google.com/covid19/mobility/data_documentation.html?hl=en#about-this-data).

**Figure A3:** Mobility evolution across Chilean regions



Note: The figure presents the different mobility indices for the Chilean regions. Each panel shows how visits and length of stay at different places changed compared to a baseline period. This baseline is the median value, for the corresponding day of the week, during the 5-week period Jan 3–Feb 6, 2020. The gray lines indicate different regions, and the black line show the national average. The vertical line indicates March 13, which denote the beginning of the COVID-19 pandemic in Chile.

**Figure A4:** Mobility evolution in Chile and the United States



Note: The figure presents national mobility indices for Chile and the United States. Each panel shows how visits and length of stay at different places changed compared to a baseline period. This baseline is the median value, for the corresponding day of the week, during the 5-week period Jan 3–Feb 6, 2020. The gray lines indicate different regions, and the black line show the national average. The vertical line indicates March 13, which denotes the beginning of the COVID-19 pandemic in Chile.

### A3 Type-specific decreases in ED visits

We quantify the drop in ED visits estimating a difference-in-differences model. For each diagnosis  $k$  we estimate the regression equation

$$Y_{dykr} = \beta_k \text{COVID-19} + \gamma_k 1\{Post\} + \mu_{dow,k} + \nu_{mk} + \tau_{yk} + \alpha_{rk} + \epsilon_{dykr} \quad (A1)$$

where  $Y_{dykr}$  is the number of visits of diagnosis  $k$  in the day  $d$  counted after January 1 of year  $y$  in region  $r$ , COVID-19 denotes the post-March 13, 2020 (pandemic) period; and  $1\{Post\}$  is an indicator for the period after March 13 in the two years in our sample. The terms  $\mu_{dow,k}$ ,  $\nu_{mk}$ ,  $\tau_{yk}$ , and  $\alpha_{rk}$  are diagnosis-dependent fixed effects for day of the week ( $dow$ ), month of the year ( $m$ ), year, and region, respectively.

We present the results in Table A1. We find a statistically significant drop in all the categories. The beginning of the COVID-19 pandemic led to a decrease of 1,430 non-respiratory emergency visits, of which 262, 25, and 138 were due to trauma or poisoning, circulatory system, and diarrhea, respectively. These figures represent a 57 drop relative to the pre-pandemic average for non-respiratory visits combined, and a 66, 38, and 92 percent drop for each respective category.

**Table A1:** Effect of the pandemic on emergency visits by category

|                    | (1)<br>All<br>Non-Respiratory | (2)<br>Trauma<br>or Poisoning | (3)<br>Circulatory<br>System | (4)<br>Diarrhea         | (5)<br>All<br>Other Causes |
|--------------------|-------------------------------|-------------------------------|------------------------------|-------------------------|----------------------------|
| COVID-19 post      | -1429.799**<br>(122.265)      | -261.807**<br>(20.882)        | -24.672**<br>(2.417)         | -138.265***<br>(10.000) | -1005.054**<br>(91.286)    |
| N                  | 3570                          | 3570                          | 3570                         | 3570                    | 3570                       |
| R-Squared          | 0.94                          | 0.92                          | 0.94                         | 0.85                    | 0.94                       |
| Mean Dep. Variable | 2,506.33                      | 396.99                        | 65.16                        | 150.75                  | 1893.43                    |

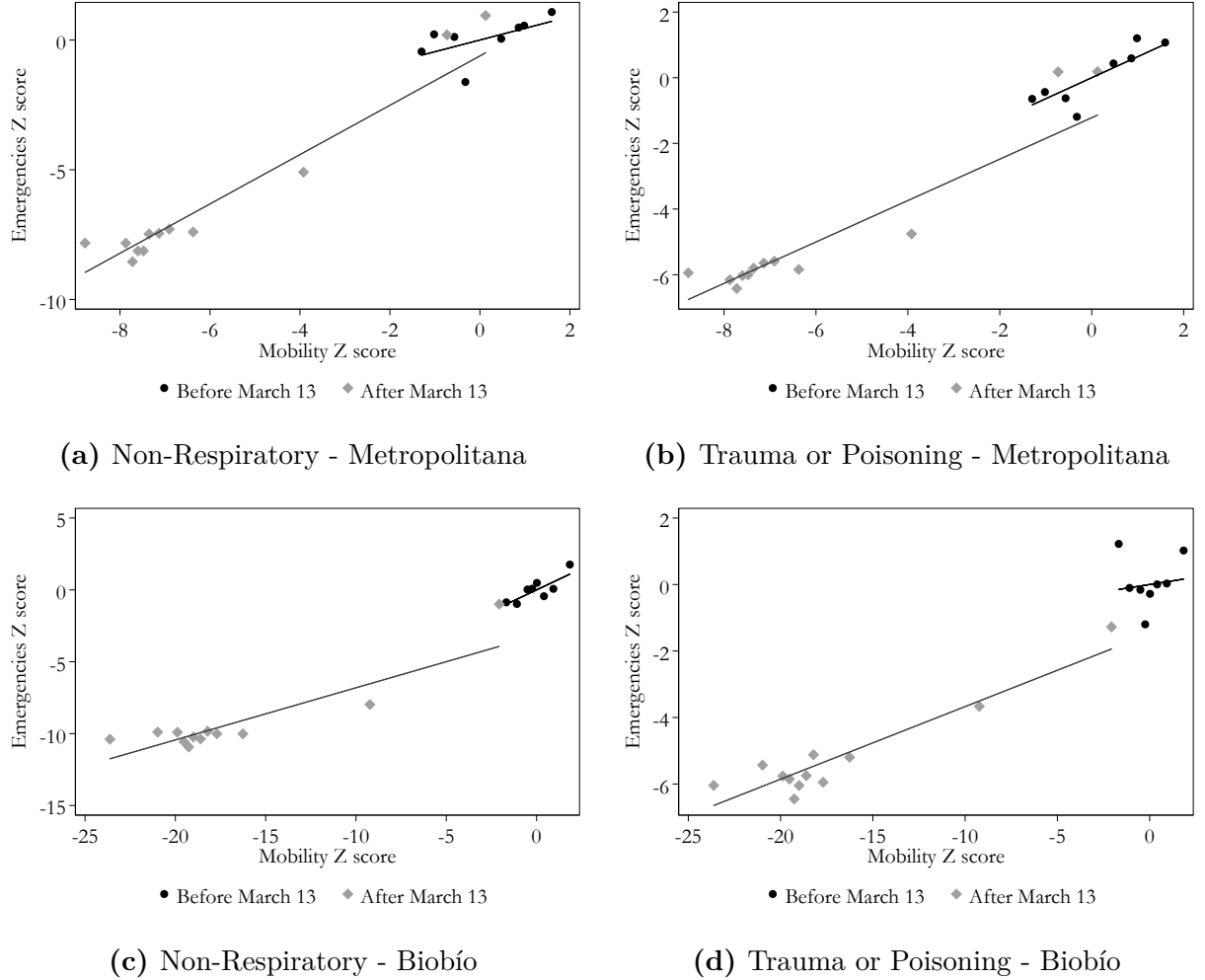
Note: The table shows the effect of the pandemic spread on total emergency room visits in Chile between January 1 and April 28, where we use 2019 as the control group for 2020. The estimation includes day of the week, month of the year, year, and regional fixed effects. The mean dependent variable includes only observations before March 13 in 2019 and 2020. Robust standard errors possibly autocorrelated up to 3 lags (Newey and West, 1987) in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

### A4 Normalized Mobility and Emergency Visits

We present in Figure A5 more examples of binned scatter plots relating (normalized) emergency visits with (normalized) mobility measures. We show these plots for all non-respiratory

conditions and for trauma and poisoning in the Metropolitana and the Biobío region, which gather 50 percent of the country's inhabitants.

**Figure A5:** Normalized Mobility and ED visits



Note: The Figure presents binned scatter plots of transit station mobility and emergency visits in the Metropolitana and Biobío regions of Chile. The dotted lines show a linear fit for observations before and after March 13, the date of the beginning of the COVID-19 pandemic.

## A5 Robustness

### A5.1 Model Specification

The following table shows the robustness of the relationship between ED visits and mobility to different model specifications.

**Table A2:** OLS estimates for various specifications

|                                   | (1)                 | (2)                 | (3)                 | (4)                | (5)                |
|-----------------------------------|---------------------|---------------------|---------------------|--------------------|--------------------|
| Residential                       | -33.98**            | -15.29              | -45.57***           | -28.49**           | -40.42**           |
| Non Residential                   | 52.18***<br>(14.49) | 62.16***<br>(16.17) | 50.28***<br>(13.24) | 72.41**<br>(32.53) | 73.17**<br>(32.87) |
| F-stat.                           | 8.85                | 8.15                | 8.26                | 5.99               | 5.90               |
| p-value                           | 0.00                | 0.00                | 0.00                | 0.00               | 0.00               |
| N                                 | 448                 | 448                 | 432                 | 448                | 448                |
| R-squared                         | 0.99                | 0.99                | 0.99                | 0.99               | 0.99               |
| Mean Dep.Variable                 | 2337                | 2337                | 2334                | 2337               | 2337               |
| Day of the week FE                | Y                   | N                   | Y                   | Y                  | Y                  |
| Excluding March 8                 | N                   | N                   | Y                   | N                  | N                  |
| Month FE                          | N                   | N                   | N                   | Y                  | Y                  |
| Month $\times$ Day of the week FE | N                   | N                   | N                   | Y                  | Y                  |

Notes: The table shows the results of a OLS estimation of Equation (1) in the pre-pandemic period. Column (1) replicates the result of the main specification presented in the text. March 8 corresponds to the International Women’s Day, where a massive march took place in many cities. Robust standard errors possibly autocorrelated up to 3 lags (Newey and West, 1987). \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

## A5.2 Principal Components Analysis

In this section, we examine the robustness of the decomposition in Table 3 to different model specifications. Each column of Table A3 presents a different specification. Columns (1)-(4) show OLS estimates and the standard errors of the explained share. Column (1) replicates the results of Table 3, where the explanatory variables related to mobility are residential and non-residential mobility. Column (2) replaces the residential and non-residential mobility indices with the two principal components that explain the most variance of the six mobility indices. Column (3) shows the results of using three principal components. Finally, Column (4) presents results of two principal components of a subset of the mobility indices workplaces, retail and recreation, and transit stations, which The Economist referred to as economic-activity index.

**Table A3:** Robustness Checks: Explained Shares (%)

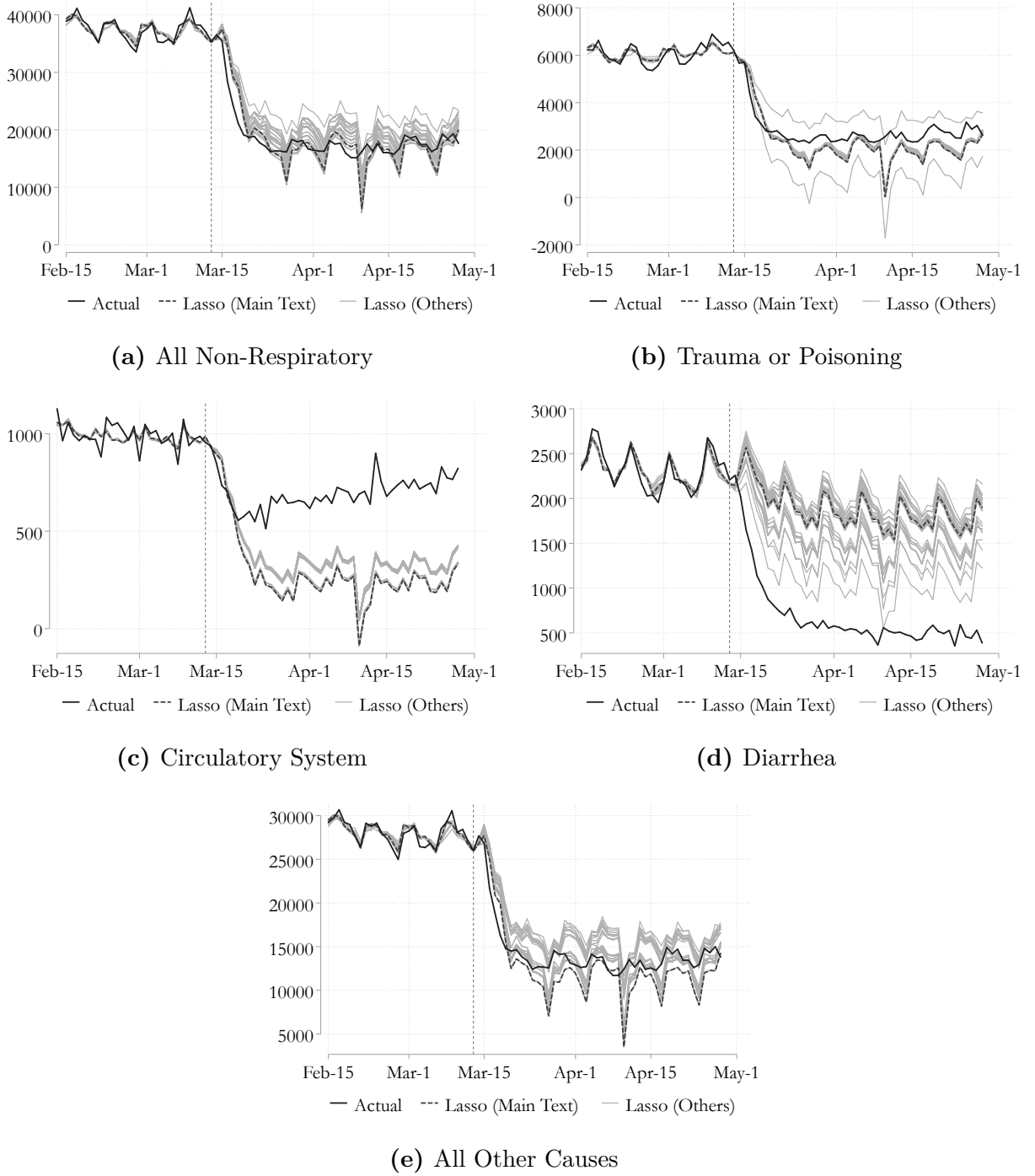
|                     | (1)              | (2)               | (3)               | (4)               |
|---------------------|------------------|-------------------|-------------------|-------------------|
| All Non-Respiratory | 52.15<br>(16.61) | 59.86<br>(22.12)  | 58.20<br>(23.45)  | 44.66<br>(25.55)  |
| Trauma or Poisoning | 90.78<br>(29.74) | 128.60<br>(39.44) | 132.04<br>(40.89) | 116.14<br>(38.31) |
| Circulatory System  | 25.98<br>(47.12) | 55.00<br>(56.29)  | 61.61<br>(57.88)  | 83.71<br>(55.12)  |
| Diarrhea            | 25.01<br>(20.54) | 20.92<br>(24.90)  | 15.00<br>(25.99)  | 0.28<br>(25.00)   |
| All Other Causes    | 46.66<br>(17.39) | 48.08<br>(22.12)  | 45.52<br>(23.59)  | 31.93<br>(26.98)  |

Notes: The table shows the ED visits explained shares in percentage terms according to different specifications as explained in the text. Robust and autocorrelated (3 lags) standard errors (Newey and West, 1987) in parentheses.

### A5.3 Lasso Regressions

Figure A6 shows plots for different seeds of the same lasso specification and the sample sample as in the main text. These plots are meant to show only the instability in variable selection in the pre-COVID period for each emergency type and not confidence intervals of lasso.

**Figure A6:** Lasso predictions—Actual and Predicted Emergencies by Type

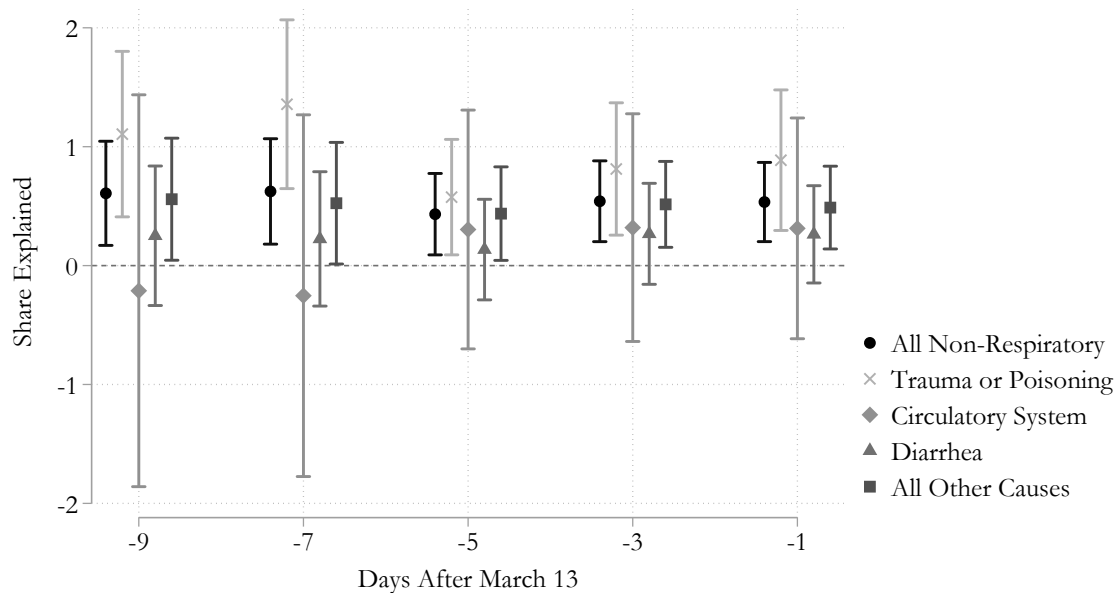


Notes: The vertical line represents the period that was used for prediction and the actual prediction using Google's Community Mobility Reports and regional fixed effects. Lasso and Elastic Net predictions are equivalent for all non-respiratory and trauma or poisoning categories.

## A5.4 Estimation period

Figure A7 shows the share of the variation explained by our OLS model by modifying the end and the beginning of the pre- and post-Covid-19 periods.

**Figure A7:** Robustness of the OLS Estimates to the Pre-Covid-19 period definition



Note: Figure shows the share of the variation explained by the model by modifying the end and the beginning of the pre- and post-Covid-19 periods. Each estimate includes 95% confidence intervals based on the standard errors of the prediction. Relative to March 13, each point estimate represent the share of the variation during the post-Covid-19 period that can be explained with pre-Covid-19 data. The end and the beginning of each period is indicated in the horizontal axis. Shares are calculated according to the main text.